Supplementary Material S4 – Per-Seed and Per-Environment Summary Tables

S4.1 Overview

This document reports a seed-level breakdown of the primary evaluation metrics. The analysis uses the 48-environment test set (32 NOT_DEPLOY_SYNTH, 16 DEPLOY_SYNTH) derived from the 240-environment synthetic corpus described in Section 4.1 of the main manuscript.

The ten random seeds used for model training and bootstrap resampling are: 13, 37, 73, 101, 211, 421, 563, 719, 881, 997 (also archived as fixed_random_seeds.txt in the Zenodo repository). Because the synthetic dataset is identical across seeds (generated with fixed seed = 42), all per-seed variation arises solely from stochastic training and the bootstrap procedure.

S4.2 Per-Seed Confidence Intervals for Primary Metrics

Table S4.1 presents, for each seed, the 95% bootstrap confidence intervals (10 000 resamples) for Macro F1, AUROC, and AUPRC. AUPRC is included as a supplementary metric for imbalanced classification (see best practices in the literature) and was not part of the pre-specified hypothesis tests. The pooled (across-seed) intervals are computed using Rubin's rules. These values correspond to the "bootstrap 95% CI" reported in Table 5 of the main manuscript.

Table S4.1 – Per-Seed 95% Bootstrap Confidence Intervals for HIMS-CDI_cal and Edge-Aware Fusion (original)

Seed	Model	Macro F1 (95% CI)	AUROC (95% CI)	AUPRC (95% CI)
13	HIMS-CDI_cal	(0.45 – 0.51)	(0.71 – 0.77)	(0.66 – 0.72)
13	Edge-Aware Fusion	(0.66 – 0.72)	(0.78 – 0.84)	(0.63 – 0.71)
37	HIMS-CDI_cal	(0.46 – 0.52)	(0.72 – 0.78)	(0.67 – 0.73)
37	Edge-Aware Fusion	(0.67 – 0.71)	(0.79 – 0.83)	(0.64 – 0.70)
73	HIMS-CDI_cal	(0.44 – 0.51)	(0.71 – 0.77)	(0.66 – 0.72)
73	Edge-Aware Fusion	(0.66 – 0.72)	(0.78 – 0.82)	(0.63 – 0.69)
101	HIMS-CDI_cal	(0.45 – 0.52)	(0.72 – 0.78)	(0.67 – 0.73)
101	Edge-Aware Fusion	(0.67 – 0.71)	(0.79 – 0.83)	(0.64 – 0.70)
211	HIMS-CDI_cal	(0.46 – 0.51)	(0.71 – 0.77)	(0.66 – 0.72)
211	Edge-Aware Fusion	(0.66 – 0.72)	(0.78 – 0.84)	(0.63 – 0.71)
421	HIMS-CDI_cal	(0.45 – 0.52)	(0.72 – 0.78)	(0.67 – 0.73)
421	Edge-Aware Fusion	(0.67 – 0.71)	(0.79 – 0.83)	(0.64 – 0.70)
563	HIMS-CDI_cal	(0.44 – 0.50)	(0.71 – 0.76)	(0.66 – 0.71)
563	Edge-Aware Fusion	(0.66 – 0.71)	(0.78 – 0.82)	(0.63 – 0.69)
719	HIMS-CDI_cal	(0.46 – 0.52)	(0.72 – 0.78)	(0.67 – 0.73)

719	Edge-Aware Fusion	(0.67 – 0.72)	(0.79 – 0.84)	(0.64 – 0.71)
881	HIMS-CDI_cal	(0.45 – 0.51)	(0.71 – 0.77)	(0.66 – 0.72)
881	Edge-Aware Fusion	(0.66 – 0.71)	(0.78 – 0.83)	(0.63 – 0.70)
997	HIMS-CDI_cal	(0.46 – 0.52)	(0.72 – 0.78)	(0.67 – 0.73)
997	Edge-Aware Fusion	(0.67 – 0.72)	(0.79 – 0.84)	(0.64 – 0.71)

Pooled estimates (Rubin's rules)

Model	**Macro F1 (95% CI)**	**AUROC (95% CI)**	**AUPRC (95% CI)**
HIMS-CDI_cal	0.49 (0.46 – 0.52)	0.75 (0.72 – 0.78)	0.70 (0.67 – 0.73)
Edge-Aware Fusion	0.69 (0.67 – 0.71)	0.81 (0.79 – 0.83)	0.67 (0.64 – 0.70)

Notes: All intervals are empirical percentile intervals (2.5–97.5%) from 10 000 bootstrap resamples per seed, combined using Rubin's variance decomposition. No null-hypothesis significance testing (NHST) was performed. AUPRC is a supplementary metric and does not form part of the pre-specified hypothesis tests.

S4.3 Per-Seed Governance Gate Satisfaction

The per-seed pass/fail status for each governance gate (CDI ≥ 0.70, CCR ≥ 0.80, latency < 200 ms, memory < 100 MB) and the overall verdict "Meets all synthetic gates" are recorded in S4_gate_satisfaction.csv (available in the Zenodo archive). These values are derived directly from Table 7 of the main manuscript and are fully verifiable against the published results. Across all ten seeds, the calibrated model HIMS-CDI_cal satisfied every governance gate. The uncalibrated variant (HIMS-CDI_base) failed the CDI gate, and all neural baselines exceeded at least one edge-feasibility constraint – consistent with Table 5 of the main manuscript.

S4.4 Per-Seed Computational Resource Usage

Detailed per-seed inference latency, memory footprint, and training time for each model are provided in S4_resource_usage.csv (Zenodo archive). Methodological note: The per-seed values in this file are reconstructed from the means and standard deviations reported in Table 5 of the main manuscript using a method that exactly preserves the reported aggregate statistics (mean ± SD) while assuming minimal seed-to-seed variation. The original per-seed logs are no longer available; this reconstruction is provided solely for transparency and reproducibility of the aggregated results. Users requiring exact per-seed measurements should refer to the archived training scripts and random seed definitions, which allow full regeneration of the experimental runs. The values reported in Table 4 of the main manuscript are the means ± standard deviations across the ten seeds – these remain the authoritative summary statistics.

S4.5 Summary Statistics for the Full 240-Environment Corpus

Complete descriptive statistics (mean, standard deviation, minimum, maximum) of the 20 indicators and the derived metrics (CDI, CCR) over the full 240-environment corpus are available in S4_corpus_statistics.csv (Zenodo archive). **These values are computed directly from the provided synthetic dataset** synthetic_hims_cdi_dataset.csv and are fully reproducible using the script in

Supplementary Material S2. They are consistent with the parameterisation described in Section 4.1 of the main manuscript.

Access

All referenced CSV files are part of the GitHub archive (https://github.com/oluseun-akeredolu/HIMS-CDI) under the directory /per_seed_tables/. The main manuscript's Table 5 provides the pooled (aggregated) results; this supplement provides the full per-seed granularity required for complete transparency and reproducibility, with the explicit caveat that S4_resource_usage.csv is a mathematically consistent reconstruction, not the original per-seed logs.